Ram Calculus and Standard-Map Metric Entropy

M281: Full-Parameter-Tail Audit, Weighted Deficit Panels, and Physical-Time Return Certificates

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Abstract

The September 23, 2026 manuscript M281 claims positive normalized-area metric entropy for every sufficiently large positive parameter of the exact standard sine map. We examine its complete mathematical source, including noncommuting transfer products, local analytic scarcity, the fast-bridge change of variables, and the weighted limiting-measure contradiction. The claimed full parameter tail is strictly stronger than the positive-measure parameter target of the relevant Ram report TR341. We do not certify the complete source proof or produce a new successful nonzero-parameter entropy certificate. Independently proved refinements include the optimal constant-one shortfall count, a sharp two-sided transfer cone, an exact rare-cancellation word, a finite-state physical-time return criterion, a fixed-parameter projective-lift variational formula, a fully explicit C^2 polynomial cap, and Ram moment bounds for the capped deficit using finite weighted measures rather than the possibly divergent underlying mass. The last construction prevents a spurious error proportional to the number of scales divided by the terminal deficit. Precise hypotheses, countercontrols, source locators, primary-literature distinctions, and remaining verification obligations accompany each conclusion.

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1 Exact Target and Evidence Standard

Let m be normalized Lebesgue area on $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, and set

$$f_k(x,y) = (x + y + k\sin(2\pi x),\ y + k\sin(2\pi x)) \pmod{\mathbb{Z}^2}. \tag{1.1}$$

The inverse lift is $(X,Y) \mapsto (X - Y, Y - k\sin(2\pi(X - Y)))$. Differentiation gives

$$Df_k(x,y) = \begin{pmatrix} 1+a & 1 \\ a & 1 \end{pmatrix}, \qquad a = 2\pi k\cos(2\pi x), \qquad \det Df_k = 1. \tag{1.2}$$

Thus f_k is an analytic area-preserving diffeomorphism. All entropy and exponent logarithms in this report are natural unless specified.

M281 [1], `sections/introduction.tex`, `main:entropy`, asserts

$$\exists k_0 > 0 \quad \forall k \geq k_0 : \quad h_m(f_k) > 0. \tag{1.3}$$

The conclusion is not merely existence of one parameter, a countable sequence of parameters, a residual parameter set, or a positive-length set of parameters. At each admissible parameter the relevant measure is the original area m, not a maximal-entropy measure, a periodic-orbit measure, or a random perturbation law. It does not assert ergodicity of the entire torus, positive exponents almost everywhere, or a common positive entropy lower bound over the tail.

For the original smooth compact map, Oseledets' theorem and the nonergodic Pesin formula give

$$\chi_k(z) = \lim_{n\to\infty} \frac{1}{n} \log \|Df_k^n(z)\| \geq 0 \quad (m\text{-a.e.}), \qquad h_m(f_k) = \int \chi_k \, \mathrm{d}m. \tag{1.4}$$

The two exponents sum to zero; bounded derivative and inverse derivative at fixed k give the required logarithmic integrability. We import the classical theorems, not prove them. The C^2 compact-manifold, smooth-area setting and the dispensability of ergodicity are also recorded in Young's Section 2, Theorem 4 [10]; M281 cites Pesin and Mañé, including the latter's erratum. Therefore $h_m(f_k) > 0$ is equivalent to $m\{\chi_k > 0\} > 0$.

Three evidence levels. *Source claim* means a theorem asserted by M281, whose argument is inspected below. *Proved here* means a displayed derivation with complete hypotheses, using explicitly named classical imports. *Conditional certificate* means the derivation is proved but its infinite-time or parameter-uniform premises have not been established for a particular nonzero standard map. Reading an entire source is not equivalent to independently certifying every analytic estimate in it. No novelty or priority claim is attached to the refinements.

Ram applicability. The catalogue's level-5 classification is warranted by the same map, measure, derivative cocycle and physical-time entropy question. It does not mean that TR341 already proves (1.3). Indeed TR341's `eq:target` asks only for positive parameter measure, and its `thm:main`, `thm:tail`, and `thm:parameters` are conditional. Its sentence disallowing an "L5 designation" on the basis of no solution is not adopted as an applicability convention here: direct relevance and completion of a solution are different judgments.

2 Primary Literature and Current Status

The bounded public check was performed on October 7, 2026, using public titles, names and identifiers only. It distinguishes the new manuscript claim from established neighboring results; an older paper's statement that the problem was open is not used to dismiss a later proof claim.

- Berger–Turaev, *On Herman's positive entropy conjecture*, Theorem A, Conjecture 0.4 and Corollary C [3]: conservative perturbations can have positive metric entropy. Their conjecture asks for a positive-measure parameter set within the exact standard family. The perturbations in their theorem need not be amplitude changes of (1.1). The introductory statements were checked, not their complete construction.

- Obata, Theorem A [4]: sufficiently large standard maps have a unique Bernoulli measure of maximal entropy. For nearby C^2 maps the uniqueness statement includes an existence qualification. The measure is not identified with area. The theorem and normalization were checked, not all invariant-manifold estimates.

- Oliveira, Theorem 5.1 and Section 5.4 [5]: positive *topological* entropy is obtained for positive parameter $\mu \neq 4$ in the amplitude convention $\mu/(2\pi)$. Here this means $k > 0$, $k \neq 2/\pi$. The

theorem and the shift-factor implication were checked; the complete prime-end argument was not audited.

- Gorodetski [6]: full Hausdorff dimension for the stochastic sea at sufficiently large topologically generic parameters is not positive area, and residual parameter sets need not have positive Lebesgue measure. The abstract and bibliographic record were checked.
- Blumenthal–Xue–Young [7]: small random perturbations give Lyapunov bounds by changing the probability law. Blumenthal's increasing-coefficient compositions [8] change the deterministic sequence of maps. Their abstracts establish these scope distinctions, not a fixed-parameter area-entropy conclusion.
- Knill's corrected version 2 [9] explicitly removes incorrect portions of an earlier entropy announcement. Its correction notice was checked. The withdrawn lower-bound claim is not an input.

M281 is a September 2026 assertion of the much stronger (1.3). The official repository states that its manuscripts are at different stages of verification and that unformalized results may contain issues [2]. This report neither assumes a historical open-problem status is unchanged nor treats repository publication as proof certification. The bounded primary-source search did not locate an independent accepted verification of this particular full-tail claim. That is a search boundary, not a theorem that none exists.

3 Complete-Source Proof Audit

3.1 Reading scope and normalization

All seven mathematical section files, the main file, bibliography, and the separate bridge figure were read. The manifest associates this download with commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; that association is manifest provenance, not an independently verified Git checkout. Exact paths and reading boundaries are in Appendix A. The following locators refer to labels in those original section files; no original source is altered.

The involution $P(x, y) = (x, x - y)$ conjugates the map to

$$q_{i+1} = 2q_i + k\sin(2\pi q_i) - q_{i-1} \pmod 1, \quad A(v_i) = \begin{pmatrix} v_i & -1 \\ 1 & 0 \end{pmatrix}, \quad v_i = 2 + K\cos(2\pi q_i), \quad K = 2\pi k. \tag{3.1}$$

The time-ordered transfer is $T_{a,b} = A(v_{b-1}) \cdots A(v_a)$ for $b > a$. In the source $M = K + 4$ and $g(i, j) = \log_M \|T_{i,j}\|$. Determinant one gives $\|T^{-1}\| = \|T\| \geq 1$ in dimension two. Consequently $0 \leq g(i, j) \leq |i - j|$, and the transfer identity gives symmetry and the triangle inequality. These are independently checked elementary steps in `setup:transfer–scale:distance-bound`. Uniform product area for (q_i, q_{i-1}) at each fixed time does not make observations at distinct times independent.

3.2 Cancellation and local analytic scarcity

The event in `can:theorem` has two length-n wings growing at least $M^{(1-\delta)n}$ but total growth at most $M^{\delta n}$. Singular input directions then yield a normalized recurrence solution small at both endpoints, `can:small-solution`. The wedge estimate $\|T_{-n,n}\| \geq D_- D_+ |a_- \wedge a_+|$ preserves the product ordering; it is not multiplication of unrelated one-step singular values. The Wronskian in

`can:wronskian` and the nonnegative shortfall count give linearly many good indices. `can:matching` pairs derivative scales within 3/5 in base-M logarithms, with exponentially many possible patterns. The final ball-cover contraction is $|V_{m+1}| \le 5\eta|V_m|$, with η chosen before the large-M threshold so that the pattern count is defeated.

The genuinely difficult local input is `can:local-scarcity`, proved in `local-scarcity.tex`. Its scope is uniform in central position, index and base configuration, with a common successful solution at each point; it is not an independence assertion. The source controls five phase derivatives, a fourth boundary-value derivative, and the Green matrix of a finite tridiagonal Dirichlet problem. The displayed exponent calculation in `loc:inverse-chains` gives $M^{-72/25}$, sufficient for the stated $M^{-2-4/5}$ bound. This arithmetic and the Wronskian mechanism were checked.

In `loc:cubic-relation`, the boundary data reduce to a cubic polynomial versus two reciprocal cosines. The proof treats both dangerous degeneracies: opposite limiting cosines are divided by $\ell = \max(\|p\|_{\mathrm{coeff}}, K^{-1})$, retaining the K^{-1} shift; a vanishing second phase slope uses a fourth-order reciprocal remainder and a quartic obstruction. The Hausdorff-limit argument selects four distinct nonpole points to bound cubic coefficients, then uses analytic identity. These steps were read through, not merely their statement. The report does not independently certify every uniform derivative constant and every subsequence alternative in this analytic package. That is an unchecked dependency, not a demonstrated counterexample.

3.3 Fast bridges without assumed mixing

In `bridge:graphs`, prefix growth at every length is used to construct finite-time contracting directions and long regular level graphs. The Green matrix and weighted contraction give a graph over a fixed coordinate interval, not only an infinitesimal tangent. The determinant-distortion identity is

$$\frac{\mathrm{d}}{\mathrm{d}r}\log|t_j(r)| = \sum_{p=0}^{j-1} t_p(r)U_p(r)^2\phi''(q_p(r)).$$

The second factor of U_p comes from differentiation of the potential; dropping it would change the summability. Noncommuting product telescoping transports endpoint observations with error $CM^{-\sigma n}$ under $3\kappa+\sigma < 0.8/3$.

`bridge:label-jacobian` and `bridge:sweep-jacobian` carry area between transverse graph families. The proof includes local smooth extensions on eligible compact subsets, so those subsets are not silently assumed to have interior. Regular-level uniqueness, small slopes and bounded projection multiplicity supply the injectivity and area factors. The conclusion is the weighted, fast-event-restricted estimate

$$\mathbb{E}[\mathbf{1}_{\{g(a,b)\ge c_b(b-a)\}}W\mathbf{1}_{\{H>h\}}] \le C(\mathbb{E}W + M^{-\sigma(b-a)})\mathbb{P}(H > h/2). \tag{3.2}$$

It concerns a bounded Lipschitz observation class and sufficiently large bridge lengths; it is not unconditional mixing of the standard map. All graph construction and change-of-variables steps were read. A full independent verification of the uniform graph constants and measurable chart assembly remains outside this report's proof certification.

3.4 Critical scales and locally finite measures

Under a hypothetical sequence $k \to \infty$ with zero exponent almost everywhere, $e_n = \mathbb{E}[1 - g(0,n)/n]$ tends to one as $n \to \infty$ at each fixed parameter, whereas $e_a \to 0$ at each fixed length as $k \to \infty$. The source correctly keeps these limits in their stated order. `scale:fourpoint` compares normalized

transfer norms with the maximum of an angle sine and two inverse-square singular values, with constant 6. Truncation plus cancellation scarcity yields $e_n \le e_{2n} \le Ce_n + C/n$. Critical scales satisfy

$$n_0 \to \infty, \quad \epsilon = e_N \to 0, \quad \epsilon n_0 \to \infty, \quad e_{2^m n_0} = o(\epsilon), \quad e_{2^m N} = O_m(\epsilon). \tag{3.3}$$

These elementary scale-selection implications and the telescoping test identities were checked.

The measures $\mu_K = \epsilon^{-1} \sum_{n_0 \le n < N}^{\text{dyadic}} \operatorname{Law}(g_n)$ and $\nu_K = \epsilon^{-1} \operatorname{Law}(g_N)$ live on distance arrays, not on phase space. Their masses need not be bounded. With affine arrays $\mathcal{A} = \{w|s-t| : 0 \le w \le 1\}$ removed, the nonnegative tests J and V detect nonaffinity and give compact-set mass bounds. The source proves vague convergence on $\mathcal{U} = X \setminus \mathcal{A}$, not weak convergence of finite probability measures. The limiting arrays are 1-Lipschitz semimetrics satisfying a zero-hyperbolic four-point inequality; no unproved real-tree representation theorem is substituted.

The dilation $Sd(s,t) = d(2s,2t)/2$ is proper on $\mathcal{U}$ because $S^{-1}\mathcal{A} = \mathcal{A}$. This justifies passage of compactly supported tests through S and the balance

$$S_*\mu = \mu + \nu, \qquad \mu = \mu_\infty + \sum_{j \ge 1} (S^{-j})_* \nu. \tag{3.4}$$

Subtraction is local on relatively compact sets, not subtraction of infinite total masses. These measure-theoretic distinctions are essential.

3.5 Shape exclusions and the final weighted balance

The cancellation estimate excludes near-total symmetric cancellation triples. The weighted bridge estimate, not a bare unweighted probability bound, excludes a fast connection from a nonaffine neighborhood to a non-unit-speed neighborhood. In `shape:classification`, the support therefore has an entirely slow region, or one or two exterior unit rays and a slow complement. `shape:crossloss` rules out a zero-length slow interval in the two-ray case and bounds crossing loss by C_*L/t, where $C_* = 2 + 6/\delta$.

The cap satisfies $\Phi(u) = \alpha u$ near zero and $\Phi(u) = 1$ for $u \ge 1 - c_b$, with $10C_*\alpha < 1$. Writing $h_t(z) = 1 - d(z, z+t)/t$, the observable is

$$G_\Phi = \Phi(h_2(0)) - \tfrac{1}{2}\Phi(h_1(0)) - \tfrac{1}{2}\Phi(h_1(1)). \tag{3.5}$$

It is signed; there is no pointwise nonpositivity claim. `end:balance` does not rely on vague convergence alone. It treats all affine speeds separately: flatness of the cap handles slow speeds; the weighted bridge eliminates intermediate speeds; truncation on a shorter dyadic interval removes a possible weighted atom at unit speed. The latter uses $\epsilon n_0 \to \infty$ to dispose of the accumulated $1/n$ error. This is the principal weighted-compactness obligation.

The resulting exact identity is $\int G_\Phi \, \mathrm{d}\mu = H_\Phi > 0$. The terminal series in (3.4) telescopes to a value strictly below H_Φ: if $\nu \ne 0$, a slow interval visible at arbitrarily fine scales has positive mass at some dyadic translate, hence at the origin by invariance. If $\nu = 0$, the series is zero. The invariant remainder is nonpositive: entirely slow arrays give zero; the one-ray class has zero locally finite translation-and-dilation invariant interface mass; two-ray bands compare fine occupancy to coarse crossing loss. The source supplies endpoint measurability, local finiteness and atomlessness for this last argument. Reading this step as a claim about an invariant phase-space probability would be wrong.

Finally, `end:contradiction` correctly uses nonnegativity in (1.4): failure of positive-area positivity at arbitrarily large parameters supplies a zero-exponent sequence. Excluding every such sequence yields an entire eventual parameter tail. No parameter-density inference or semicontinuity argument is needed for this final logical step.

3.6 Audit disposition

No explicit mathematical counterexample to a source assertion is established in this report. The elementary transfer, scale, normalization, telescoping and final-quantifier checks described above are positive checks at their stated depth. The complete uniform local scarcity theorem, fast-bridge construction and all passages combining them are not certified here as one independently established chain. Accordingly, (1.3) remains a *source claim under review* in this report, not a theorem obtained by the Ram representation. Section 10 isolates what a quantitative replacement of the final balance would have to prove. An independent source review, October 7, 2026, likewise found no concrete error in its bounded full-source checks, without certifying the complete proof. Its actual scope and local evidence file are recorded in the appendix.

4 A Sharp Shortfall Counting Refinement

The source's `setup:shortfall-count` uses disjoint intervals and their threefold enlargements, giving the numerical constant 4 in its displayed proof. The ordered endpoint structure permits an exact improvement, independently of any entropy hypothesis.

Theorem 4.1 (Sharp one-dimensional shortfall count)**.** *Let $z_0, \dots, z_N \in \mathbb{R}$, with $z_i - z_{i-1} \le 1$. Set $d_i = 1 - z_i + z_{i-1} \ge 0$ and $D = \sum_{i=1}^{N} d_i$. For $c < 1$ let $h = 1 - c > 0$. Call $j \in \{1, \dots, N\}$ backward-bad if some integer $0 \le l < j$ satisfies $z_j - z_l < c(j - l)$. Then the number B_- of backward-bad endpoints satisfies*

$$B_- \le D/h, \qquad B_- < D/h \quad \text{if } B_- > 0. \tag{4.1}$$

The identical bound holds for forward-bad starts $j \in \{0, \dots, N-1\}$ admitting $j < l \le N$ with $z_l - z_j < c(l - j)$. The universal multiplicative constant 1 is sharp.

Proof. Backward badness means that a witnessing step interval $(l, j]$ has $\sum_{i=l+1}^{j} d_i > h(j - l)$. Choose the largest remaining bad endpoint j, choose any one of its witnesses, and remove every remaining bad endpoint in $(l, j]$. At most $j - l$ bad endpoints are removed. Every remaining bad endpoint is at most l, so all subsequently selected witness step intervals are disjoint from this one. Repeat until there are no bad endpoints. Finiteness guarantees termination. Summing over the disjoint witnesses gives

$$B_- \le \sum_{\text{selected}} (j - l) < h^{-1} \sum_{\text{selected}} \sum_{i=l+1}^{j} d_i \le D/h$$

when at least one witness was selected. If the bad set is empty, its count is zero and the non-strict inequality holds directly. For forward badness choose the smallest remaining bad start and remove starts in $[j, l)$, using the same step set $\{j+1, \dots, l\}$. Subsequent starts are at least l, giving disjoint step sets again.

For sharpness take $d_1 = hN + \varepsilon$, $d_i = 0$ for $i > 1$, $\varepsilon > 0$, and recover z_i by the prescribed increments. Every endpoint $1 \le j \le N$ is backward-bad using $l = 0$, so $B_- = N$ while $D/h = N + \varepsilon/h$. Letting $\varepsilon \downarrow 0$ rules out every smaller universal constant. Reversing the deficit sequence supplies the forward example. □

The strict inequality has the integer consequence $B_- \le \lceil D/h \rceil - 1$ for a nonempty bad set. The direct form (4.1) is often more convenient. Replacing the source's covering estimate by this theorem reduces that contribution to the bad-index count in `can:many-good` and to the removed-middle-index

count in `bridge:middle-eligibility`. Two orientations still require their respective contributions, and neighboring increment tests still need their stated multiplicities. Other unspecified constants and the local analytic thresholds remain; no computable k_0 follows.

5 Concrete Ram Representation and Ordered Products

5.1 Actual local inputs

TR341 [11] was read in full. Its nonergodic return theorem supplies the exact conditional connection between coherent return expansion and area entropy. TR248's `thm:perturb` and `eq:decodepe rt` supply physical-coordinate inversion error [12], including operator error multiplied by signal amplitude. The revised foundation's `eq:moments` and "A concrete polynomial Ram construction" supply unisolvent weighted polynomial moments with full column rank [13]. Volume 21's physical-metric, interval, target-observable and parameter sections were read directly. They supply finite norms, inclusion and error accounting [14], not recurrence or a positive-area theorem. Exact files are listed in the appendix.

5.2 From measured moments to an actual function

On a compact interval or parameter panel, let ϕ be a column of N linearly independent polynomial functions, with physical Gram matrix $G = \int \phi\phi^T \,\mathrm{d}\omega \succ 0$. A Ram bank of bounded functionals has matrix $E_{ij} = \ell_i(\phi_j)$. Assume *full column rank* and a positive data metric W. For an actual scalar coefficient function $v = \phi^T c + r$, suppose $\|r\|_\infty \le \eta$ and measured data obey $y = Ec + e$, $\|W^{1/2}e\|_2 \le \varepsilon$. The error e must include the moments of the actual remainder r, as well as acquisition error. Define

$$\alpha = \inf_{\|x\|_2=1} \|W^{1/2}EG^{-1/2}x\|_2 > 0, \quad \widehat{c} = (E^TWE)^{-1}E^TWy, \quad K_0 = \sup_z \sqrt{\phi(z)^TG^{-1}\phi(z)}. \tag{5.1}$$

Whitening by $G^{1/2}$ and least squares give

$$\|v - \phi^T\widehat{c}\|_\infty \le \eta + K_0\varepsilon/\alpha. \tag{5.2}$$

Indeed the whitened coefficient error is at most ε/α, and physical Cauchy–Schwarz bounds evaluation by K_0 times that error. This is a finite-model theorem, not recovery of an arbitrary continuum function from a finite bank. Bounds for derivatives require corresponding remainder and evaluation norms, not a new name for the same samples.

For $v_i = 2 + K\cos(2\pi q_i)$, an enclosure from (5.2) gives an actual transfer-factor error because

$$\|A(v_i) - A(\widehat{v}_i)\|_2 = |v_i - \widehat{v}_i|. \tag{5.3}$$

An error in the orbit must first enter the scalar bound. For a lifted approximate recurrence with residual τ_i, parameter error $|k - k_*| \le \kappa$ and $|k|, |k_*| \le K_*/(2\pi)$, valid bounds are

$$\begin{aligned} e_{i+1} &\le (K_* + 4)e_i + \kappa + \tau_i, \\ |v_i - \widehat{v}_i| &\le 2\pi\kappa + 2\pi K_* e_i + \eta_i, \end{aligned} \tag{5.4}$$

where e_i bounds Euclidean pair error and η_i bounds evaluation error at the approximate pair for k_*. The first line follows from the global transfer derivative bound and the parameter forcing $(k - k_*)\sin(2\pi q)$ in the first component. The second uses the cosine Lipschitz constant. Consistent lifts or interval unions must cover torus crossings. No long-time shadowing is inferred.

Proposition 5.1 (Noncommuting product enclosure)**.** *Suppose* $\|A_i - \widehat{A}_i\| \le \delta_i$, $\|\widehat{A}_i\| \le b_i$. *For* $T = A_{n-1}\cdots A_0$ *and* $\widehat{T} = \widehat{A}_{n-1}\cdots\widehat{A}_0$,

$$\|T - \widehat{T}\| \le \Delta_n := \sum_{i=0}^{n-1} \delta_i \prod_{j=i+1}^{n-1} (b_j + \delta_j) \prod_{j=0}^{i-1} b_j. \tag{5.5}$$

If $b_i \le B$ *and* $\delta_i \le d$, *then* $\Delta_n \le (B+d)^n - B^n$. *Empty products equal one.*

Proof. Subtract the products by replacing factors successively. The exact difference is $\sum_i A_{n-1}\cdots A_{i+1}(A_i - \widehat{A}_i)\widehat{A}_{i-1}\cdots\widehat{A}_0$. Take norms, use $\|A_j\| \le b_j + \delta_j$, and sum the scalar geometric telescoping expression in the uniform case. No factors commute. □

For an exact determinant-one T, the interval

$$\log\max(1, \|\widehat{T}\| - \Delta_n) \le \log\|T\| \le \log(\|\widehat{T}\| + \Delta_n) \tag{5.6}$$

is valid when the stated norms and errors themselves are certified. Its endpoints require directed transcendental enclosures in an implementation. The upper endpoint is at least zero by consistency. Division by $n \log M$ gives a finite source-shortfall enclosure.

For nonorthogonal tangent syntheses J_x, J_y with Gram matrices $G_x, G_y \succ 0$, the physical norm of a coefficient map B is $\|G_y^{1/2} B G_x^{-1/2}\|_2$. Intermediate coordinate changes must cancel in an ordered product. The fixed matrix of P above has condition number $(3+\sqrt{5})/2$, so conjugacy changes finite logarithmic norms by at most its logarithm, but changes no Lyapunov exponent. The analysis may consistently use the constant transfer-coordinate tangent norm throughout; it may not change the norm adversarially at each time without accounting for that change.

5.3 Finite-time averages are not lower entropy bounds

For either one fixed smooth tangent norm, put $U_n(k) = n^{-1}\int \log\|Df_k^n\|\,\mathrm{d}m$. Subadditivity and invariance give nU_n a subadditive integral sequence. The usual division $n = qm + r$ proves $\lim U_n = \inf_n U_n$. Dominated convergence in (1.4) identifies this infimum with $h_m(f_k)$. Each U_n is continuous in k on compact parameter intervals because its integrand is continuous and uniformly bounded there. Thus

$$h_m(f_k) = \inf_{n\ge1} U_n(k), \qquad \{k : h_m(f_k) > 0\} = \bigcup_{j\ge1} \{k : h_m(f_k) \ge 1/j\} \tag{5.7}$$

is an F_σ set, and entropy is upper semicontinuous. Positive finite-time lower bounds, even for many times, do not bound this infimum from below. At $k = 0$, $Df_0^n = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$ has positive log norm at every finite time and zero limiting exponent. Upper semicontinuity does not make a positive value persist in a parameter neighborhood.

6 Sharp Transfer Cones and an Exact Cancellation Control

Proposition 6.1 (Two-sided hyperbolic strip)**.** *For* $B > 2$, *put*

$$\lambda_B = \frac{B + \sqrt{B^2 - 4}}{2}, \qquad r_B = \lambda_B^{-1}, \qquad C_B = \{(x,y) \ne 0 : |y| \le r_B|x|\}. \tag{6.1}$$

Every $A(v)$ *with* $|v| \ge B$ *preserves this double cone and expands every cone vector in Euclidean norm by at least* $\lambda_B > 1$. *The common lower factor is sharp. Independently, the full operator norm and conorm are*

$$\sigma(v) = \frac{\sqrt{v^2+4} + |v|}{2}, \qquad \sigma(v)^{-1}. \tag{6.2}$$

Proof. For a cone vector write $s = y/x$, $|s| \le r_B$. The new slope has absolute value $|v - s|^{-1} \le (B - r_B)^{-1} = r_B$. Its squared norm ratio is $((v - s)^2 + 1)/(1 + s^2)$. With $t = |s|$, this is at least $((B - t)^2 + 1)/(1 + t^2)$. The difference of the numerator and $\lambda_B^2(1 + t^2)$ is a concave quadratic in $t \in [0, r_B]$. At r_B it is zero; at zero it is $B^2 + 1 - \lambda_B^2 = 3 + \lambda_B^{-2} > 0$. It is therefore nonnegative throughout. Equality occurs at $v = B$, $s = r_B$, proving sharpness. Finally $A(v)^T A(v)$ has trace $v^2 + 2$ and determinant one. Its singular values are the two positive numbers in (6.2) and their squares have that trace and product, proving the formula. □

For the actual standard map, let $K = 2\pi k > 0$, and clamp a real number to $[-1, 1]$ by $[u]_{-1}^{1} = \max(-1, \min(1, u))$. The bad strip $\{|2 + K\cos(2\pi q)| < B\}$ has area

$$p_B(K) = \frac{1}{\pi}\left[\arccos\left[\frac{-B-2}{K}\right]_{-1}^{1} - \arccos\left[\frac{B-2}{K}\right]_{-1}^{1}\right]. \tag{6.3}$$

This is the elementary distribution of cosine under uniform q; endpoint equalities have zero area. Stationarity and a union bound, without independence, show that the probability of n consecutive good sites is at least $(1 - np_B(K))_+$. In transfer coordinates,

$$\frac{1}{n}\mathbb{E}\log\|T_{0,n}\| \ge (1 - np_B(K))_+ \log\lambda_B. \tag{6.4}$$

On the good event a single cone vector is expanded at every step; on its complement determinant one supplies the lower bound zero. For fixed n, choosing $B = M^{1-\zeta}$ recovers the source's fixed-time large-parameter growth, with $\log\lambda_B/\log M \to 1 - \zeta$. For fixed K the bound becomes uninformative as $n \to \infty$. It is an actual finite-time result, not a positive-entropy certificate.

Proposition 6.2 (Rare cancellation in the exact transfer alphabet)**.** *Let* $J = A(0) = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. *For all real* v *and integers* $n \ge 0$,

$$A(v)JA(-v) = -J, \qquad A(v)^n JA(-v)^n = (-1)^n J. \tag{6.5}$$

Consequently periodic ordered words of length $2n + 1$, *containing* n *copies each of* $A(v)$ *and* $A(-v)$ *and one* J *in the displayed product order, have zero top exponent. For* $|v| > 2$, *the fraction of sites not covered by Proposition 6.1 tends to zero as* $n \to \infty$.

Proof. Direct multiplication proves the first identity. Surround the induction hypothesis on the left by $A(v)$ and on the right by $A(-v)$ to prove the second. The full-period product is an orthogonal quarter-turn up to sign and its powers have norm one. For each fixed word, the finitely many prefixes have a finite maximum norm; products of any number of periods followed by a prefix are bounded. Their logarithmic growth per step tends to zero. Exactly one factor per period is J. □

This is a cocycle countercontrol in the *same matrix alphabet*, not a claim that its coefficient sequence satisfies the nonlinear standard-map recurrence for one k and one orbit. Even when the scalar values lie in $[2 - K, 2 + K]$, realizability is an additional condition. It refutes the proposed general inference "rare bad factors plus strong good factors imply a positive exponent." Scalar conorm costs do not restore the direction destroyed at a bad return.

7 All-Return Growth in Physical Time

7.1 Nonergodic theorem and complete tail cost

Let f be a C^2 diffeomorphism of a compact surface preserving a normalized smooth area m. Assume the entropy formula and bounded derivative/inverse logarithms as above. Let A be a measurable

base of positive area, r its first return time, $F = f^r$ its return map, and $S = \bigcup_{j\in\mathbb{Z}} f^j A$ its saturation, all modulo recurrence null sets. The levels $f^\ell\{r = n\}$, $0 \le \ell < n$, partition S. Indeed overlapping levels would give an earlier return; the last visit before a given point selects its unique level. Applying invariance and countable additivity gives

$$\int_A r \,\mathrm{d}m = m(S), \qquad \int_S \chi \,\mathrm{d}m = \int_A r\chi \,\mathrm{d}m. \tag{7.1}$$

The return map preserves $m_A = m|_A/m(A)$, since the images of its return branches partition A. In particular $r \in L^1(m_A)$, with mean $m(S)/m(A)$, not necessarily $1/m(A)$.

Theorem 7.1 (Coherent return expansion)**.** *Suppose a measurable cone field $C(x) \subset T_xM \setminus \{0\}$, nonempty for almost every $x \in A$, and a real $a \in L^1(A, m_A)$ obey, for almost every $x \in A$ and every $v \in C(x)$,*

$$Df^{r(x)}v \in C(Fx), \qquad \|Df^{r(x)}v\| \ge e^{a(x)}\|v\|. \tag{7.2}$$

Then, writing $\bar a = \mathbb{E}_{m_A}(a \mid \mathcal{I}_F)$ and $\bar r = \mathbb{E}_{m_A}(r \mid \mathcal{I}_F)$,

$$\bar r\chi \ge \bar a \quad a.e., \qquad h_m(f) \ge \int_A (\bar a)_+ \,\mathrm{d}m \ge \max(0, \int_A a \,\mathrm{d}m). \tag{7.3}$$

Proof. Remove all backward F-iterates of the exceptional set and intersect with the Oseledets and Birkhoff full-measure sets. Follow one nonzero vector from its initial cone through every return. With $R_n = \sum_{j<n} r(F^j x)$, the ordered chain rule gives $\log \|Df^{R_n}(x)\| \ge \sum_{j<n} a(F^j x)$. Birkhoff gives $R_n/n \to \bar r \in [1, \infty)$ and the gain average converges to $\bar a$. Oseledets along this subsequence proves the first inequality. Since χ is bounded, nonnegative and F-invariant, (7.1) gives $\int_S \chi = \int_A \bar r\chi \ge \int_A (\bar a)_+$. Use Pesin for the original smooth f and the integral preservation of conditional expectation. No differentiability of the measurable return map or ergodicity of the base is assumed. □

This reproduces TR341's `thm:main` with its actual hypotheses. Clipping a before conditional averaging is invalid. Alternating gains $1, -1$ on a two-cycle have conditional mean zero but positive integral of a_+. An invariant block may be separated; arbitrary good/bad cells are not invariant blocks.

If $\|Df^{\pm 1}\| \le e^L$, short-return cells have valid gains ℓ_j and area intervals $[p_j^-, p_j^+]$, and $T_R \ge \int_{A\cap\{r>R\}} r \,\mathrm{d}m$, then the all-return cone condition *still required on the tail* gives

$$h_m(f) \ge \max\left(0, \sum_{\ell_j \ge 0} \ell_j p_j^- + \sum_{\ell_j < 0} \ell_j p_j^+ - LT_R\right). \tag{7.4}$$

This follows by setting $a = -Lr$ on the tail and applying Theorem 7.1. The inverse bound proves a conorm estimate, not cone inclusion. With $p_n = m(A \cap \{r > n\})$, Tonelli gives exactly $\int_{r>R} r \,\mathrm{d}m = Rp_R + \sum_{n=R}^\infty p_n$. Small unweighted omitted area does not replace this time-weighted cost.

7.2 A finite-state refinement with cycle ratios

Theorem 7.2 (Return-graph physical-time certificate)**.** *In the preceding setting suppose a finite measurable partition $A = \bigsqcup_{i=1}^s A_i$ and nonempty cone fields $C(x) \subset T_xM \setminus \{0\}$ satisfy (7.2)'s inclusion for every actual return. Suppose every return from A_i to A_j has a label $e : i \to j$ in a finite directed graph, and throughout that class*

$$r(x) \le t_e < \infty, \qquad \|Df^{r(x)}v\| \ge e^{\ell_e}\|v\| \quad (v \in C(x)).$$

Labels may include unrealized edges, but no realized return may be omitted. For a number $\gamma \geq 0$, if every directed cycle obeys

$$\sum_{e \text{ in cycle}} \ell_e \geq \gamma \sum_{e \text{ in cycle}} t_e, \tag{7.5}$$

then $\chi \geq \gamma$ almost everywhere on S and $h_m(f) \geq \gamma m(S) \geq \gamma m(A)$. The largest value supported by these finite inequalities is $\min_{cycles} \sum \ell_e / \sum t_e$, when this ratio is nonnegative, the graph has a cycle, and the chosen t_e are positive. If the minimum ratio is negative, this test supplies no admissible nonnegative γ.

Proof. Set $w_e = \ell_e - \gamma t_e$. The cycle condition says that there are no negative directed cycles. Add a vertex with zero edges to all vertices, and let p_i be the minimum path weight to vertex i. Removing cycles never increases a path weight, so a minimum exists among finitely many simple paths. Therefore $p_j \leq p_i + w_e$, or $\ell_e \geq \gamma t_e + p_j - p_i \geq \gamma r(x) + p_j - p_i$. Following one cone vector through n returns gives

$$\log \|Df^{R_n}(x)\| \geq \gamma R_n + p_{i_n} - p_{i_0}.$$

The potential is bounded on a finite set and $R_n \to \infty$. Oseledets gives $\chi \geq \gamma$ on A, then on its saturation by invariance; Pesin gives the integral bound. Conversely summing a potential inequality around a cycle yields (7.5). Every closed walk decomposes into cycles, so checking simple cycles suffices; taking the minimum ratio proves the final assertion. □

The bounded-return hypothesis in this finite version is strong and is not asserted for the standard map. An unbounded-return extension is valid if each tail class satisfies directly

$$\log \frac{\|Df^{r(x)}v\|}{\|v\|} \geq \gamma r(x) + p_j - p_i \quad \text{and preserves the actual endpoint cone} \tag{7.6}$$

for *all* its returns. The same telescoping proof applies. It cannot follow merely from $\|Df^{-r}\| \leq e^{Lr}$ when $\gamma > 0$. In applications, one may apply the theorem on a proved return-invariant subbase of positive area while using (7.4) elsewhere.

Exact calibration. A two-state graph with forced alternating edges of gains $\ell_{12} = 2$, $\ell_{21} = -1/2$ and time bounds $t_{12} = t_{21} = 1$ has cycle ratio $3/4$. Choosing $p_1 = 0$, $p_2 = 5/4$ gives equality in both edge inequalities for $\gamma = 3/4$. Alternating diagonal matrices $\mathrm{diag}(e^2, e^{-2})$ and $\mathrm{diag}(e^{-1/2}, e^{1/2})$ on the horizontal cone ray realize this abstract gain calculation. The negative edge is retained, not discarded. Adding a self-loop of gain $-1/2$ defeats the positive cycle certificate. These are exact graph and cocycle fixtures, not standard-map return partitions.

7.3 Finite matrix gates and parameter transport

If $\|B - B_0\| \leq \delta$, and for every source-cone unit vector $\|B_0 v\| \geq s > \delta$ and $\mathrm{dist}(B_0 v, C^c_{\mathrm{target}}) \geq \rho > \delta$, then $Bv \in C_{\mathrm{target}}$ and $\|Bv\| \geq s - \delta$. This is the triangle inequality and the definition of distance to the complement. Combine (5.2) and (5.5) to obtain the physical δ. The target is the cone at the *actual* return point; endpoint uncertainty and varying cone directions must be covered.

If the same complete dynamical graph and cones are valid throughout a parameter interval, and certified bounds give $\ell_e(k) \geq \ell_{e0} - d_e$, $t_e(k) \leq t_e^+$, then replacing cycle weights by $\ell_{e0} - d_e - \gamma t_e^+$ is a valid uniform test. Alternatively, fixed potentials with center slack $s_e = \ell_{e0} - \gamma t_{e0} - p_j + p_i > 0$ remain valid whenever $|\ell_e(k) - \ell_{e0}| + \gamma |t_e(k) - t_{e0}| < s_e$. These bounds do not prove constancy of the return partition, all-return cone persistence or survival tails. Those premises must hold independently at every parameter. A finite sweep proves none of them by itself.

8 A Direction-Retaining Projective-Lift Criterion

There is also an exact probability-measure counterpart that retains the ordered tangent direction, rather than multiplying separate matrix norms. Fix one parameter k throughout this section. On the compact space $X = \mathbb{T}^2 \times \mathbb{RP}^1$ use the original flat physical tangent norm and define

$$\widehat{f}(z,[u]) = (f_k(z), [Df_k(z)u]), \qquad g(z,[u]) = \log \frac{\|Df_k(z)u\|}{\|u\|}. \tag{8.1}$$

The derivative is invertible, so both objects are continuous on the whole projective space and g is bounded. Let $\pi : X \to \mathbb{T}^2$ be the base projection, and write $\mathcal{P}_m$ for the invariant Borel probabilities η satisfying $\widehat{f}_*\eta = \eta$ and $\pi_*\eta = m$.

Theorem 8.1 (Fixed-parameter variational formula)**.** *With the smooth-area hypotheses of* (1.4)*,*

$$h_m(f_k) = \max_{\eta \in \mathcal{P}_m} \int_X g \, \mathrm{d}\eta. \tag{8.2}$$

Moreover, suppose a sequence of Borel probabilities η_j on X has $\pi_*\eta_j \rightharpoonup m$, invariance residuals

$$\int (\psi \circ \widehat{f} - \psi) \, \mathrm{d}\eta_j \longrightarrow 0 \quad \textit{for every continuous } \psi, \tag{8.3}$$

and $\liminf_j \int g \, \mathrm{d}\eta_j \geq b > 0$. Then $h_m(f_k) \geq b$. A fixed uniformly dense countable family of continuous tests suffices in (8.3)*.*

Proof. The chain rule and cancellation of intermediate vector norms give

$$\sum_{i=0}^{n-1} g \circ \widehat{f}^i(z,[u]) = \log \frac{\|Df_k^n(z)u\|}{\|u\|}.$$

If $\eta \in \mathcal{P}_m$, integration yields $n \int g \, \mathrm{d}\eta \leq \int \log \|Df_k^n\| \, \mathrm{d}m = nU_n(k)$. Taking the infimum in (5.7) proves the upper bound.

For the converse, at each z choose whichever of the two physical orthonormal coordinate vectors has larger image under $Df_k^n(z)$, using a fixed tie rule. This is a Borel choice $u_n(z)$ and

$$\|Df_k^n(z)\|/\sqrt{2} \leq \|Df_k^n(z)u_n(z)\| \leq \|Df_k^n(z)\|.$$

The lower bound follows from the sum of the two squared column norms bounding the squared operator norm. Lift m to the graph of $[u_n(z)]$, obtaining ζ_n, and set $\eta_n = n^{-1} \sum_{i=0}^{n-1} \widehat{f}^i_* \zeta_n$. Each η_n projects to m and, for continuous ψ,

$$\left| \int (\psi \circ \widehat{f} - \psi) \, \mathrm{d}\eta_n \right| \leq 2\|\psi\|_\infty / n, \qquad U_n(k) - \frac{\log 2}{2n} \leq \int g \, \mathrm{d}\eta_n \leq U_n(k).$$

The first estimate telescopes; the second uses the chain-rule identity. Weak compactness of probabilities on compact X gives a convergent subsequence. Its limit has marginal m and is invariant by continuity of the tests. Continuity of g and $U_n(k) \to h_m(f_k)$ show that the limit attains the asserted maximum.

For the conditional statement, take a weakly convergent subsequence of the proposed η_j. The marginal and residual hypotheses give an element of $\mathcal{P}_m$, and continuity gives its gain at least b. For a dense test family, uniform approximation of ψ incurs at most twice its uniform approximation error because each measure is a probability. Thus all continuous tests have vanishing residuals as well. □

Concrete operator and Ram moment form. The Koopman operator $\mathcal{U}\psi = \psi \circ \widehat{f}$ acts on continuous functions with $\|\mathcal{U}\| = 1$. Invariance is precisely $\eta((\mathcal{U} - I)\psi) = 0$. Finite Ram moment banks can represent the actual marginal tests, these residuals, and the gain observable. If ψ is replaced by p with $\|\psi - p\|_\infty \le \delta$, the residual changes by at most 2δ; if g is replaced by q with $\|g - q\|_\infty \le \varepsilon$, its mean changes by at most ε. Data error must be added as in (9.8) below. Positivity and unit mass of the represented measures, the correct marginal, and residual control on increasingly rich tests are essential. A single finite moment bank does not imply these limiting hypotheses or a nonvanishing gain margin.

In transfer coordinates, away from the projective chart poles, $t_{i+1} = 1/(v_i - t_i)$ and the exact gain is

$$\log \frac{\|T_{0,n}(1, t_0)\|}{\|(1, t_0)\|} = \sum_{i=0}^{n-1} \log |v_i - t_i| + \frac{1}{2} \log \frac{1 + t_n^2}{1 + t_0^2}. \tag{8.4}$$

This follows by writing $A(v_i)(1, t_i) = (v_i - t_i)(1, t_{i+1})$. It is not valid across a chart pole without changing charts; the global projective formulation (8.1) has no such defect.

Changing to a continuous uniformly positive definite physical Gram metric changes g by $\log t \circ \widehat{f} - \log t$, where t is the ratio of the two norms on the line. This bounded continuous coboundary integrates to zero under every invariant lift. Arbitrary coefficient rescaling without transporting the metric is not such a legitimate substitution.

Scope and failure controls. The variational identity is proved, but no lift with positive limiting gain is constructed here for any nonzero parameter. At $k = 0$ the finite-horizon lifts in the proof can have positive gain and residual $O(1/n)$ while their gain tends to zero. Thus one nearly invariant finite lift is insufficient. The source's locally finite array measures μ, ν cannot serve as η: their spaces, normalization and invariances differ. Nor can fixed-k weak compactness be silently applied as $k \to \infty$, when both the map and the observable vary. This supplies a precise direction-retaining formulation, not a proof of the entire parameter tail.

9 An Explicit Cap and Weighted Ram Deficit Panels

9.1 A polynomial replacement for the unspecified smooth cap

Proposition 9.1 (Exact C^2 cap)**.** *Let $0 < a < 1$, $b = a/4$, $L = a - b$, and $\alpha > 0$ satisfy $\alpha a < 1$. Define*

$$C = 30\left(\frac{1 - \alpha b}{L} - \frac{\alpha}{2}\right) > 0, \qquad t = \frac{u - b}{L}.$$

The following piecewise polynomial is C^2 and nondecreasing on $[0, 1]$, with $0 \le \Phi \le 1$:

$$\Phi(u) = \begin{cases} \alpha u, & 0 \le u \le b, \\ \alpha b + L\left[\alpha(t - t^3 + \frac{1}{2}t^4) + C(\frac{1}{3}t^3 - \frac{1}{2}t^4 + \frac{1}{5}t^5)\right], & b \le u \le a, \\ 1, & a \le u \le 1. \end{cases} \tag{9.1}$$

It has positive linear upper and lower comparison constants with u. For any source constants $c_b < 1$ and $C_* > 0$, choose rational $0 < a \le 1 - c_b$ and rational $0 < \alpha < \min(1/a, 1/(10C_*))$. Then this cap has every regularity and flatness property used in M281's final balance.

Proof. The derivative on the middle panel is

$$p(t) = \alpha(1 - 3t^2 + 2t^3) + Ct^2(1 - t)^2 = \alpha(1 - t)^2(1 + 2t) + Ct^2(1 - t)^2 \ge 0.$$

It has $p(0)=\alpha$, $p(1)=0$ and $p'(0)=p'(1)=0$. Its integral is $\alpha/2+C/30$; the definition of C makes the middle endpoint value equal to one. Thus values and the first two derivatives match. Positivity of C follows from $\alpha(b+L/2)<\alpha a<1$. Monotonicity proves the range bound. The ratio $\Phi(u)/u$ extends continuously to $\alpha>0$ at zero and is positive on the compact interval, proving its positive minimum and finite maximum. Extend linearly to the left and constantly to the right. The source uses only bounded first and second derivatives, their continuity at the plateau, exact linearity near zero, and the prescribed small α. Hence C^2 suffices for the cited balance arguments; no higher cap derivative is invoked there. □

This supplies a finite, exact Ram panel representation: three polynomial pieces of degrees one, five, and zero with explicit interface jets. For rational a,α, all polynomial coefficients are rational. Moments against $1,u,\dots,u^5$ with positive weight are unisolvent on the middle panel by positive definiteness of their Gram matrix. No fitted transition or unproved cap shape is needed. The source's c_b,δ and a valid parameter threshold are still not numerically supplied by this construction. A globally analytic polynomial cannot have both nontrivial linear and constant open pieces; retaining panels is mathematically necessary for these exact properties.

9.2 A factorization by the two controlled defects

Put $a_1=h_1(0)$, $a_2=h_1(1)$, $\bar a=(a_1+a_2)/2$, $d=(a_1-a_2)/2$, $J=h_2(0)-\bar a\ge 0$, and $V=(a_1-a_2)^2$. For sampled or supported arrays, $a_1,a_2,h_2(0)\in[0,1]$.

Theorem 9.2 (Weighted observable identity and stability)**.** *For any C^2 cap Φ, define*

$$A_\Phi(a_1,a_2,J)=\int_0^1 \Phi'(\bar a+tJ)\,\mathrm{d}t, \tag{9.2}$$
$$B_\Phi(a_1,a_2)=-\frac{1}{8}\int_0^1 (1-t)[\Phi''(\bar a+td)+\Phi''(\bar a-td)]\,\mathrm{d}t.$$

Then

$$G_\Phi=JA_\Phi+VB_\Phi,\qquad |A_\Phi|\le\|\Phi'\|_\infty,\quad |B_\Phi|\le\|\Phi''\|_\infty/8. \tag{9.3}$$

If a positive measure μ satisfies $L_J=\int J\,\mathrm{d}\mu<\infty$, $L_V=\int V\,\mathrm{d}\mu<\infty$, then for any second C^2 function Ψ,

$$\left|\int (G_\Phi-G_\Psi)\,\mathrm{d}\mu\right|\le\|\Phi'-\Psi'\|_\infty L_J+\frac{1}{8}\|\Phi''-\Psi''\|_\infty L_V. \tag{9.4}$$

No finiteness assumption on $\mu(X)$ is required.

Proof. The fundamental theorem of calculus gives $\Phi(\bar a+J)-\Phi(\bar a)=JA_\Phi$. Taylor's formula with integral remainder at $\bar a$, applied to $\bar a\pm d$, gives

$$\tfrac12[\Phi(\bar a+d)+\Phi(\bar a-d)]-\Phi(\bar a)=\frac{d^2}{2}\int_0^1(1-t)[\Phi''(\bar a+td)+\Phi''(\bar a-td)]\,\mathrm{d}t.$$

Since $d^2=V/4$, subtraction proves (9.3). The integral of the two weights $(1-t)$ is one, proving the bounds. Apply the same identity to $\Phi-\Psi$ and integrate against the two finite positive weighted measures $J\mu$ and $V\mu$. □

For the source's prelimit μ_K, the relevant masses have exact uniform bounds already at each aligned finite scale sum. Put $b_q = \mathbb{E}[(1 - g(0,q)/q)^2]$. With $J = J([0,2])$ and $V = V([0,2])$,

$$\begin{aligned} \int J \,\mathrm{d}\mu_K &= 1 - \frac{e_{n_0}}{\epsilon} \le 1, \\ \int V \,\mathrm{d}\mu_K &\le 8 - \frac{4e_{n_0}}{\epsilon} - \frac{4b_N}{\epsilon} \le 8. \end{aligned} \tag{9.5}$$

For completeness, write the two adjacent length-q shortfalls as X, Y and the length-$2q$ shortfall as Z. Then $0 \le X, Y, Z \le 1$, $\bar{X} = (X+Y)/2 \le Z$, $J = Z - \bar{X}$, and $V = (X-Y)^2$. Therefore

$$V/4 = (X^2 + Y^2)/2 - \bar{X}^2 \le (X^2 + Y^2)/2 - Z^2 + 2J.$$

Stationarity, dyadic summation and $\epsilon = e_N$ give

$$\frac{1}{4} \int V \,\mathrm{d}\mu_K \le \frac{b_{n_0} - b_N}{\epsilon} + 2\left(1 - \frac{e_{n_0}}{\epsilon}\right).$$

Using $b_{n_0} \le e_{n_0}$ proves (9.5). The J identity is the corresponding first-moment telescope. This sharpens the loose asymptotic bound by retaining the signs in `scale:local-mass`'s second-moment argument. Consequently

$$\left| \int (G_\Phi - G_\Psi) \,\mathrm{d}\mu_K \right| \le \|\Phi' - \Psi'\|_\infty + \|\Phi'' - \Psi''\|_\infty. \tag{9.6}$$

These bounds pass to nonnegative weighted vague limits by compact cutoffs, but do not prove tightness near the removed affine arrays. An unstructured uniform error η on G would instead give only $\eta\mu_K(X) = \eta\epsilon^{-1}\log_2(N/n_0)$, which can diverge. The weighted factorization is a concrete improvement in the correct observable norm, not a proof that the limiting dynamical measures exist.

9.3 Certified moment evaluation on finite weighted laws

Push $J\mu_K$ forward by (a_1, a_2, J) to a finite measure ρ_J on $[0,1]^3$, and push $V\mu_K$ forward by (a_1, a_2) to ρ_V on $[0,1]^2$. The actual support of ρ_J also has $\bar{a} + J \le 1$. Extend (9.1) constantly above one; the formulas (9.2) then define continuous functions on both whole cubes. We obtain the exact representation

$$\int G_\Phi \,\mathrm{d}\mu_K = \int A_\Phi \,\mathrm{d}\rho_J + \int B_\Phi \,\mathrm{d}\rho_V. \tag{9.7}$$

For polynomial approximants $P = \sum c_\beta x^\beta$ and $Q = \sum d_\beta x^\beta$, suppose $\|A_\Phi - P\|_\infty \le \eta_J$, $\|B_\Phi - Q\|_\infty \le \eta_V$ and actual moment enclosures are $|m_\beta^J - \widehat{m}_\beta^J| \le \varepsilon_\beta^J$, $|m_\beta^V - \widehat{m}_\beta^V| \le \varepsilon_\beta^V$. Then the Ram moment estimator $\widehat{I} = \sum c_\beta \widehat{m}_\beta^J + \sum d_\beta \widehat{m}_\beta^V$ obeys

$$|\widehat{I} - \int G_\Phi \,\mathrm{d}\mu_K| \le \eta_J L_J + \eta_V L_V + \sum |c_\beta| \varepsilon_\beta^J + \sum |d_\beta| \varepsilon_\beta^V. \tag{9.8}$$

This follows by integration of the uniform approximation errors and the triangle inequality for the moment data. At the aligned source scales, the first two terms are at most $\eta_J + 8\eta_V$ by (9.5). The moments are of the actual deterministic orbit law with its weights, not a product law postulated from separate marginals. Statistical estimates without rigorous deterministic enclosures do not meet these premises.

There is an explicit approximation rate. The cap in (9.1) has finite $K_2 = \|\Phi''\|_\infty$ and a global Lipschitz constant K_3 for Φ'', obtained from the maximum absolute third derivative of its pieces.

Differentiating the integral for A_Φ bounds its three partial derivatives by $K_2/2$; thus it is $\sqrt{3}K_2/2$-Lipschitz. Each convex-combination argument of Φ'' in B_Φ is 1-Lipschitz in (a_1, a_2), giving constant $K_3/8$. Tensor Bernstein polynomials of degree m in each coordinate therefore provide

$$\eta_J \le \frac{3K_2}{4\sqrt{m}}, \qquad \eta_V \le \frac{\sqrt{2}K_3}{16\sqrt{m}}. \tag{9.9}$$

To verify the rate, the tensor Bernstein value is the expectation of the function at independent binomial grid coordinates X_i/m with means x_i and variances at most $1/(4m)$. Cauchy–Schwarz bounds $\mathbb{E}\|X/m - x\|_2 \le \sqrt{d/(4m)}$; apply the displayed Lipschitz constants for $d = 3, 2$. This auxiliary binomial identity is a polynomial approximation tool, not an independence claim about the dynamics.

The coefficients at rational grids can be integrated from the piecewise polynomial cap with exact interval splits. Bernstein-basis moments may be used directly in (9.8), replacing monomial coefficients and errors by those for the chosen basis. Conditioning of the conversion and moment acquisition costs remain real. No moments of an unknown limiting source law were numerically computed here.

10 A Defect Budget for the Limiting Contradiction

The observable bounds above make finite representation errors explicit, but they do not remove the source's compactness obligations. For example, on $X = [0, 1]$, with the removed set $\{0\}$, let $\mu_n = n\delta_{1/n}$, $J(x) = G(x) = x$ and $V = 0$. Then μ_n tends vaguely to zero on $(0, 1]$ while $\int G \, \mathrm{d}\mu_n = 1$. Even domination $|G| \le J$ and a uniform bound on $\int J$ do not prove disappearance of weighted mass at the removed set. This is an exact measure-theoretic counterexample to a shortcut, not a counterexample to M281, which separately handles that mass.

Proposition 10.1 (Robust contradiction ledger)**.** *Suppose a proposed limiting argument has finite real numbers H, I, T, R, nonnegative e, ρ, and $D > 0$, with*

$$|I - H| \le e, \qquad I = T + R, \qquad T \le H - D, \qquad R \le \rho. \tag{10.1}$$

Then $D \le e + \rho$. In particular a proved $D > e + \rho$ rules out the proposed zero-exponent sequence.

Proof. The hypotheses give $H - e \le I = T + R \le H - D + \rho$. Rearrangement proves the assertion. All quantities must be well-defined finite integrals or convergent series before this algebra is applied. □

For M281, $I = \int G_\Phi \, \mathrm{d}\mu$, T is the backward-dilate terminal series, and $R = \int G_\Phi \, \mathrm{d}\mu_\infty$. The source claims $e = 0$ after all affine-speed checks and $\rho = 0$. It obtains a strictly positive terminal loss D by a countable dyadic-translation argument, not a numerical lower bound. If $\nu = 0$, one can instead use $D = H > 0$. These statements, with (3.4), are enough for its contradiction.

A numerical refinement must charge quadrature, moment, cap and limit errors to the correct side of (10.1). The weighted estimate (9.8) can control a finite integral error; it does not control escape to affine arrays or certify a uniform $D > 0$. The source's selected n_0, N, ϵ and its local-scarcity compactness are noneffective as presented. A computable threshold k_0, a common positive entropy lower bound, and termination of a certificate search do not follow merely from their existence arguments.

11 Parameter Scope, Advantages, and Research Priorities

11.1 Three different routes

A successful instance of Theorem 7.1 proves one parameter result. Uniform verification over a positive-length interval, with positive-area bases and all-return hypotheses at each parameter, proves the weaker positive-parameter-measure target. To prove (1.3) by certificates one needs a family covering every $k \geq k_0$, for example a proved countable cover by certified intervals with no gaps. A finite collection of bounded intervals cannot do this. An analytic parameter-uniform theorem valid above a threshold could.

The source instead uses exclusion of every unbounded zero-exponent sequence. That logical route is appropriate to the full tail and need not provide an effective uniform atlas or a computable stopping rule. Requiring such effectivity would strengthen its claim unnecessarily. Conversely an isolated numerical interval, or even an unbounded set of successful intervals with gaps, cannot be substituted for its conclusion.

Even positivity on a whole compact parameter interval plus upper semicontinuity does not imply a strictly positive common lower bound: the upper semicontinuous function with $h(0) = 1$ and $h(x) = x$ on $0 < x \leq 1$ is a counterexample to that general inference. This is a topological countercontrol, not an asserted entropy function. Bounds from the return graph or signed budget can supply a common positive margin only when their quantified premises actually hold.

11.2 Concrete benefits and costs

The transfer cone uses both signs of v and has an exact optimal common expansion factor for the declared cone. The cancellation word proves that sparse critical visits cannot be discarded at the matrix level. The finite-state potential retains negative returns and certifies a physical-time exponent without an assumed frequency law. Ram moment reconstruction makes the scalar coefficient and product error explicit, while the piecewise polynomial cap and J, V factorization target the source's actual limiting observable. These are mathematically justified local refinements, not generic transform analogies.

The costs remain substantial: product and orbit errors can grow exponentially in horizon; near-critical returns can erase angular margins; positive-area all-return bases are not supplied; graph overapproximation can introduce artificial adverse cycles. High-degree moment inversion and basis conversion can be ill-conditioned. A failed certificate is not evidence of zero entropy unless an actual counterexample to its mathematical target has been proved.

11.3 Prioritized next mathematical work

1. Verify the uniform constants and all normalization cases of `can:local-scarcity`, including the opposite-cosine and vanishing-slope limits, in an independently organized proof.

2. Certify the finite-Dirichlet graph extent, distortion and eligible subset change of variables in `bridge:graphs` and `bridge:theorem`, with a dependency list for every threshold.

3. Recheck the three affine-speed ranges in `end:balance` and the two-ray band summation in `end:remainder` as an independent measure-theoretic argument. The defect ledger distinguishes a missing limit proof from an ordinary finite arithmetic error.

4. For a constructive alternative, propose one actual positive-area base, prove return coverage and endpoint cones including every tail return, and then apply the graph or signed-area certificate.

This is the missing dynamical construction, not a choice of polynomial basis.

5. Develop rigorously enclosed weighted joint moments for (9.7); compare direct observable approximation with full density reconstruction. Any claimed parameter range must include whole-cell and whole-tail variation, not only derivative samples.

12 Conclusion

M281 claims area metric entropy positivity on the entire sufficiently large parameter tail. The inspected argument addresses the relevant noncommuting, positive-area and limiting-weight obstructions rather than replacing them by topological entropy or finite-time stretching. Its full analytic chain is not certified by this report. The Ram corpus is directly applicable to the same core problem, but supplies conditional return and representation results, not the missing global proof.

The proved contributions here include the optimal constant-one shortfall count, exact finite-matrix and observable identities, the fixed-parameter projective-lift variational formula, conditional physical-time certificates, and quantitative representation-error refinements with explicit boundaries. No successful nonzero-parameter entropy experiment, new numerical k_0, or independently completed proof of (1.3) is claimed.

A Exact Evidence and Reproducibility Locators

A.1 Pinned manuscript

Project root:

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math
```

The manifest `_sources/manuscripts.json`, id 281, and catalogue `00-Applicability-Catalogue/applicability.json`, id 281, were read. The complete mathematical source root is

```
_sources/openai-math/preprints/Positive-Metric-Entropy-for-the-Standard-Map-at-Large-Parameters-September-23-2026/build
```

Read in full: `paper.tex`, `references.tex`, `figures/bridge-sweep.tex`, and the seven files `sections/introduction.tex`, `sections/setup.tex`, `sections/cancellation.tex`, `sections/local-scarcity.tex`, `sections/bridge.tex`, `sections/scales.tex`, and `sections/geometry.tex`. Vendored code is not an input to the audit. Reading covers all mathematical derivations; independent verification depth is delimited in Section 3 rather than inferred from file coverage.

A.2 Exact local Ram files

1. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-341-astra.tex`. Read in full. Main relevant locators: `thm:main`, `lem:tower`, `thm:tail`, `prop:products`, `prop:orbit`, `thm:parameters`, `prop:paramerror`, `prop:upper`.

2. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/TechReps/FinalTechReps/TRs/TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex`. Read lines 301–431, including the interval-average fixture and E2 implementation-stable frame certificates. Locators: `thm:perturb`, `eq:decodepert`, and physical-coordinate discussion.

3. C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex. Read the representation subsection at eq:moments, lines 155–179, and the concrete polynomial construction, lines 455–474. Full-column-rank, physical-metric and finite-model restrictions are retained.

4. Under C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/, read in full ch01d-physical-metrics.tex, ch02/recurrence.tex, ch09/intervals.tex, ch10a-observables.tex, and ch10e-parameters.tex. Key locators are v21c01:physical-gram, v21c01:full-residual, v21c09:containment, v21c10:targetlemma, and v21c10:paramlemma. The finite moment recurrence is not a Poincare return theorem and is not used as one. The observable section motivates direct weighted-target error accounting; the estimates in this report are proved explicitly.

No corpus map is used as a substitute for an actual mathematical source. Original sources remain unchanged. Confidential local material has not been submitted to public search; literature queries use public metadata.

A.3 Verification boundaries

The independent source review is _audit/M281-source-audit.txt, dated October 7, 2026. It records a complete mathematical-source reading, detailed independent checks, residual validation limits, and SHA256 hashes of all source inputs. Its Section 6.4 supplies the projective-lift construction rederived in Section 8. Its Section 11 records a bounded follow-up on the report's local results, including the nonzero-cone hypothesis clarification and the improved finite bound (9.5). Neither that follow-up nor this report is a certification of the entire source theorem.

Exact arithmetic fixtures accompany the report locally in _tools/check_m281.py, with results in _audit/M281-local-checks.json. The 7,993 successful checks cover finite transfer identities, a rational cone specialization, the graph potential, cap interface jets and polynomial moments, and the weighted observable identity, including explicit failure controls. Separate results in _audit/check_batch3_parent.json record 117,180 exact integer forward/backward shortfall cover checks and 60 sharpness checks. The endpoint-greedy proof, not finite enumeration, establishes Theorem 4.1. Such checks do not prove the infinite-time standard-map premises or substitute for the mathematical proofs above. Public reading scopes are stated in Section 2; no full audit of the external classical ergodic theory or neighboring papers is claimed.

References

[1] OpenAI, *Positive Metric Entropy for the Standard Map at Large Parameters*, September 23, 2026, pinned manuscript, commit adc7f1241b42e322a6451854ab7e4b4c146bf78a. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Positive-Metric-Entropy-for-the-Standard-Map-at-Large-Parameters-September-23-2026. Full mathematical source read; certification scope in Section 3.

[2] OpenAI, *Math repository README*, verification status notice, checked October 7, 2026. https://github.com/openai/math#readme.

[3] P. Berger and D. Turaev, *On Herman's positive entropy conjecture*, Advances in Mathematics 349 (2019), 1234–1288. https://doi.org/10.1016/j.aim.2019.04.002. Theorem A, Conjecture 0.4 and Corollary C checked in https://arxiv.org/html/1704.02473v1.

[4] D. Obata, *Uniqueness of the measure of maximal entropy for the standard map*, Commentarii Mathematici Helvetici 96 (2021), 79–111. https://doi.org/10.4171/CMH/508. Theorem A and normalization checked in https://arxiv.org/html/2003.00236v1.

[5] F. Oliveira, *Positive topological entropy for the Standard Map*, arXiv:2404.02209v2 (2024), Theorem 5.1, Section 5.4. https://arxiv.org/html/2404.02209v2.

[6] A. Gorodetski, *On stochastic sea of the standard map*, Communications in Mathematical Physics 309 (2012), 155–192. https://doi.org/10.1007/s00220-011-1365-z. Abstract checked at https://arxiv.org/abs/1007.1474.

[7] A. Blumenthal, J. Xue and L.-S. Young, *Lyapunov exponents for random perturbations of some area-preserving maps including the standard map*, Annals of Mathematics 185 (2017), 285–310. Publisher abstract checked: https://annals.math.princeton.edu/2017/185-1/p05.

[8] A. Blumenthal, *Statistical properties for compositions of standard maps with increasing coefficient*, Ergodic Theory and Dynamical Systems 41 (2021), 981–1024. https://doi.org/10.1017/etds.2019.115. Abstract checked at https://arxiv.org/abs/1710.09058.

[9] O. Knill, *The Problem of Positive Kolmogorov–Sinai entropy for the Standard map*, arXiv:math/9908014v2, November 25, 2005. Correction notice and version record only: https://arxiv.org/abs/math/9908014v2.

[10] L.-S. Young, *Entropy in Dynamical Systems*, survey, Section 2, setting and Theorem 4. https://math.nyu.edu/~lsy/papers/entropy.pdf. Statement used for the smooth-area entropy formula; not a proof audit.

[11] M. Subbarao, *Exact standard-map entropy: physical first-return cones, complete tail costs, and conditional parameter certificates*, local TR-item-341-astra, September 26, 2026. Exact path and labels in Appendix A.

[12] M. Subbarao, *Stable Adaptive Ram Frames*, local TR248, E2 implementation-stable frame certificates. Exact read range in Appendix A.

[13] M. Subbarao, revised Ram foundation report, `TR-rev-1-old-tr250`, representation subsection and concrete polynomial construction. Exact path in Appendix A.

[14] M. Subbarao, Ram Calculus book corpus, volume 21, physical metrics, recurrence, intervals, target observables and parameter certificates. Exact files in Appendix A.